

Environment Monitoring System Using Arduino



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A comfortable environment can increase the productivity multi-folds. So it is important that the environment variables, such as temperature, relative humidity, dew point, light intensity and air quality (gas/smoke), are continuously monitored and corresponding systems adjusted to maintain a comfortable working environment.

Presented here is an Arduino-based hand-held environment monitoring system that senses these parameters and shows on an LCD display as shown in Fig. 1. All the mentioned parameters are displayed in a cyclic manner.

Circuit and working

Fig. 2 shows the circuit of the environment monitoring system. The heart of the system is Arduino Uno board with

microcontroller ATmega 328. Other components used are regulator 7805 (IC1), serial 16×2 LCD (LCD1), light-dependent resistor NORP12 (LDR1), gas sensor MQ135 (GS1), temperature sensor LM35 (IC2), humidity sensor SY-HS-220 (SENS1) and a few discrete components.

Arduino UNO. Arduino UNO is an open source prototyping platform based on ATmega328 microcontroller. It consists of 14 digital input/output (I/O) pins, six analogue inputs, a USB connection for programming the on-board microcontroller, a power jack, an ICSP header and a reset button. It is operated with a 16MHz crystal oscillator and contains everything needed to support the microcontroller.

Temperature sensor (LM35). It is a 3-pin temperature sensor IC that measures temperature in degree Cen-

tigrade and gives output response of 10 mV/°C. Its response is linear and highly suitable for interfacing with the analogue-to-digital converter (ADC) of any microcontroller. For example, if temperature is 25°C, then its output is 250 mV. The sensor operates off 5V DC supply. The output of the temperature sensor is connected to analogue input A5 of the Arduino board.

Humidity sensor (SY-HS-220). This humidity sensor converts the

Test Points

Test point	Details
TP0	0 V (GND)
TP1	5 V
TP2	Output of air quality sensor
TP3	Output of light sensor
TP4	Output of humidity sensor
TP5	Output of temperature sensor



Fig. 1: Parameters indicated on the display

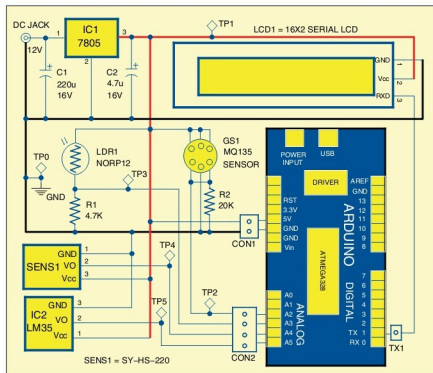


Fig. 2: Circuit of environment monitoring system